EDA Report: Geldium Delinquency Risk Analysis

> **Project:** AI-Driven Delinquency Risk Assessment for Geldium's Credit Card Collections
> **Document 1 of 4:** Exploratory Data Analysis (EDA)

Project Context

Geldium is a credit card lender looking to reduce the number of customers who miss payments (delinquency). The goal of this project is to build an **AI-powered risk assessment tool** that helps the Collections team identify at-risk customers early, prioritise outreach, and reduce delinquency in a **fair and transparent** manner.

This first document covers the exploratory analysis of Geldium's customer dataset: data quality issues, how they were treated, and the early risk signals that shape the predictive model.

Step 1: Key Insights from Initial Data Review

1.1 Notable Missing or Inconsistent Data

Field	Issue Observed	Notes
`monthly_income`	~18% missing values	Concentrated among younger customers and those new to the credit system
`credit_utilization`	Values exceeding 1.5	Suggests data entry errors or inconsistent reporting (utilisation above 100% is not realistic)
`last_payment_date`	Inconsistent formats and missing entries	Missing entries occur mostly for customers currently classified as delinquent

1.2 Key Anomalies

- **Many loans, low debt burden:** Several customers show **more than 10 active loans** yet a **low debt-to-income ratio** — an unrealistic combination that needs verification.
- **High balance, zero payment:** A subset of users has **high credit card balances but zero recorded monthly payments**, pointing to possible data gaps or pipeline errors.

1.3 Early Indicators of Delinquency Risk

- **High credit utilisation (above 90%)** is strongly associated with late or missed payments.
- **Missing or irregular** `payment_history_3m` **data** is often seen in customers who have been delinquent for **more than 60 days**.
- A **debt-to-income ratio above 0.6** frequently appears among customers with recent delinquencies.

1.4 Summary of Initial Observations

The dataset contains several informative features for modelling delinquency risk — credit utilisation, payment history, and income-related ratios. However, three problems must be addressed before modelling:

1. Missing income data
2. Inconsistent formatting (especially dates)
3. Outliers (especially in credit utilisation)

These findings form the foundation for cleaning the dataset and designing the predictive model.

Step 2: Missing Data Treatment Strategy

Feature	Identified Issue	Handling Method	Justification
`monthly_income`	Missing in ~18% of entries	Impute using the **median by age group**	Median values account for demographic variation in income and are robust to skew
`credit_utilization`	Outliers and nulls (~10%)	**Cap at 1.0**; impute nulls using the **mean**	Utilisation above 100% is unrealistic; the mean preserves the overall trend
`last_payment_date`	Inconsistent format and missing values	**Standardise the format**; impute with the **mode**	Ensures uniformity and reflects typical customer behaviour

Why this matters: Income is a core input to the debt-to-income ratio, and utilisation is the single strongest early risk signal. Treating them carefully prevents the model from learning from data errors rather than real behaviour.

Step 3: Detected Risk Patterns and Factors

3.1 High-Risk Indicators

#	Risk Indicator	What It Signals
1	**High Credit Utilisation (>90%)**	Potential overextension; a strong signal of repayment difficulty
2	**Frequent Missed Payments in the Last 3 Months**	A behavioural trend that often precedes future delinquency
3	**High Debt-to-Income Ratio (>0.6)**	Financial pressure and limited repayment capacity
4	**Recent Employment Changes**	Income instability, which may increase default risk
5	**No Auto-Pay Enrollment**	Non-enrollment is correlated with a higher likelihood of missed payments

3.2 Additional Insights

- **First-time credit users** show elevated risk because they lack historical behaviour data — but they also **respond better to early-stage interventions**.
- **Older customers (above 55)** generally show **lower delinquency rates** despite moderate outstanding balances, possibly reflecting stronger financial discipline.

Final Summary

The exploratory analysis of Geldium's customer dataset reveals clear risk factors for credit card delinquency. **Credit utilisation, recent payment behaviour, and income-to-debt ratios** show significant predictive value.

The dataset is broadly suitable for modelling, but targeted imputation and cleaning are needed — particularly in the **income** and **payment history** fields.

These insights feed directly into the design of an AI-powered risk assessment tool that helps Geldium **prioritise outreach** and **reduce delinquency** fairly and transparently.

Key takeaways

- ~18% of income values are missing → median imputation by age group
- Utilisation outliers (>1.5) → capped at 1.0
- Top early warning signs: utilisation >90%, recent missed payments, DTI >0.6
- Employment changes and lack of auto-pay add further risk
- Customers over 55 are lower risk; first-time users are higher risk but highly responsive to early outreach